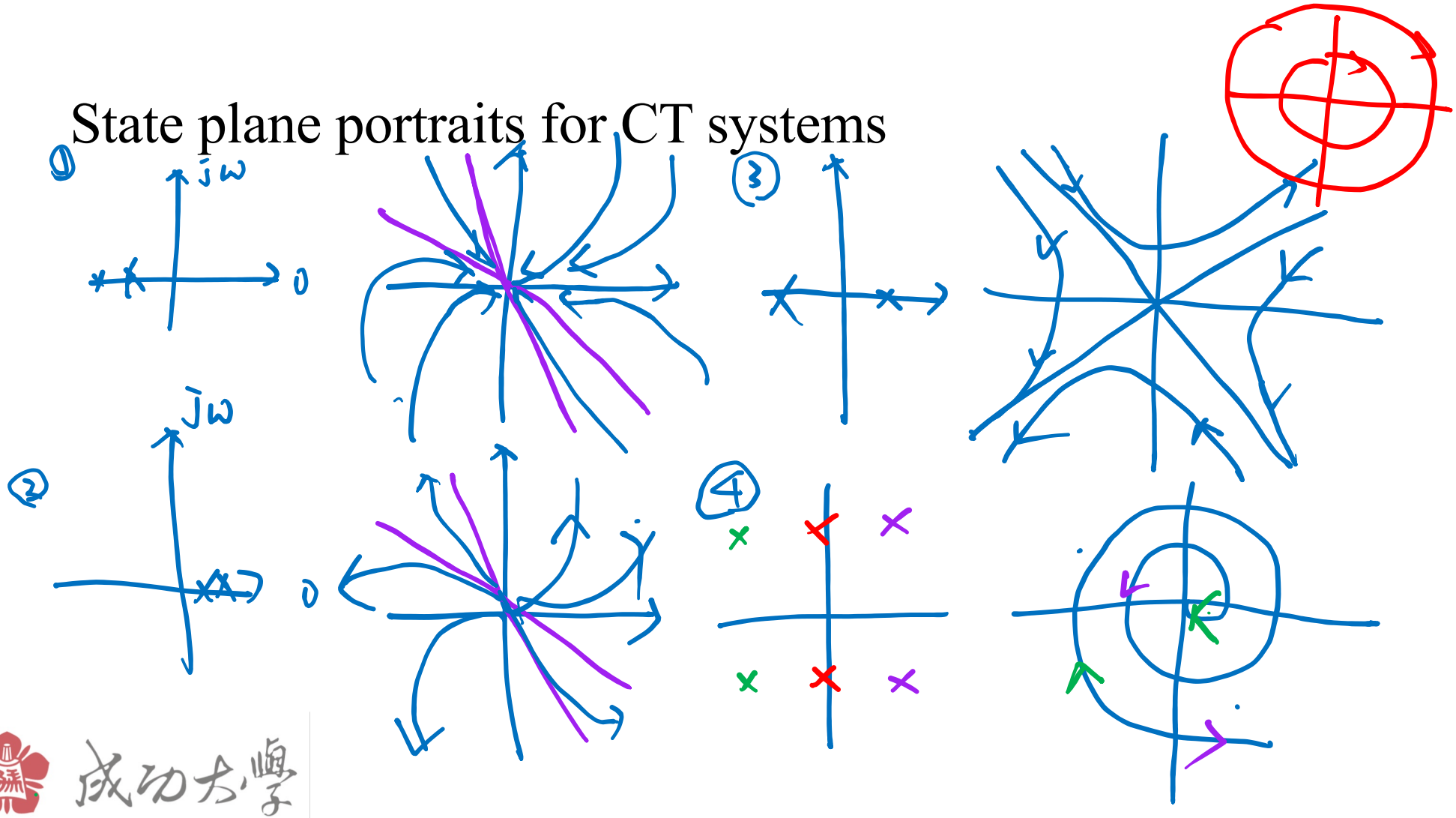
Linear System Theory

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State plane portraits for CT systems



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Polynomials of a square matrix

$s^0 = 1 \Rightarrow A^0 = I$

- Consider any polynomial p of a scalar variable s given by:

$$p(s) = a_0 + a_1 s + a_2 s^2 + \ldots + a_m s^m$$

If A is a square matrix, we define p(A) as

$$p(A) = a_0 I + a_1 A + a_0 A^2 + \ldots + a_m A^m$$

we replace s^k with A^k

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a) The matrix A & $p(A)$ commute: $A \cdot p(A) = p(A) \cdot A$

b) Consider any nonsingular matrix T, suppose $A = T \Lambda T^{-1}$

$$p(A) = T\, p(\Lambda)\, T^{-1}$$

c) Suppose A is diagonal $A = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}$

$$p(A) = \begin{bmatrix} p(\lambda_1) & & & \\ & p(\lambda_2) & & \\ & & \ddots & \\ & & & p(\lambda_n) \end{bmatrix}$$

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(d) Suppose A is diagonalizable, for some nonsingular T

$$A = T \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} T^{-1}$$

$$p(A) = T \begin{bmatrix} p(\lambda_1) & & & \\ & p(\lambda_2) & & \\ & & \ddots & \\ & & & p(\lambda_n) \end{bmatrix} T^{-1}$$

(e) If λ is an eigenvalue of A w/ eigenvector is v

$\Rightarrow p(\lambda)$ is an eigenvalue of $p(A)$ w/ eigenvector is v

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* characteristic poly of A
$p(\lambda) = \det(\lambda I - A)$

Example 85 For

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

→ 2個 eigenvalues.

we found that

$$T = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = (v_1 \quad v_2)$$

results in

$$\Lambda = T^{-1}AT = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

Hence, $A = T\Lambda T^{-1}$ and

$$p(A) = T\, p(\Lambda)\, T^{-1}$$

$$= T \begin{pmatrix} p(2) & \\ & p(4) \end{pmatrix} T^{-1}$$

$$= \frac{1}{2} \begin{pmatrix} p(2)+p(4) & p(2)-p(4) \\ p(2)-p(4) & p(2)+p(4) \end{pmatrix}$$

* $p(\cdot)$ is characteristic polynomial of A.
$p(2) = p(4) = 0 \Rightarrow p(A) = 0$

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Cayley-Hamilton Theorem

Theorem 3 *If p is the characteristic polynomial of a square matrix A, then*

$$\boxed{p(A) = 0}$$

proof. A is diagonalizable

diagonal

$\Lambda = T^{-1}AT \Rightarrow A = T\Lambda T^{-1}$

A has a matrix is similar to diagonal matrix

Ldi $p(A) = T\,p(\Lambda)\,T^{-1}$

$$= T \begin{bmatrix} p(\lambda_1) & & & \\ & p(\lambda_2) & & \\ & & \ddots & \\ & & & p(\lambda_n) \end{bmatrix} T^{-1}$$

$\lambda_i \Rightarrow$ eigenvalues of A

$$= T \begin{bmatrix} 0 & & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{bmatrix} T^{-1} = 0$$

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Cayley-Hamilton Theorem

Suppose the characteristic polynomial of A is

$$p(s) = \det(sI - A) = a_0 + a_1 s + \ldots + a_{n-1}s^{n-1} + s^n$$

$$p(A) = a_0 I + a_1 A + \cdots + a_{n-1} A^{n-1} + A^n$$

Cayley-Hamilton $\Rightarrow$ $p(A) = 0$

$$\Rightarrow \quad A^n = -a_0 I - a_1 A - \cdots - a_n A^{n-1}$$

A^m can be expressed as a linear combination of $A^0, A^1, \cdots, A^{n-1}$ ($m \geq n$)

($A^0 = I$)

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Example 87 To illustrate the result above, consider the companion matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$p(s) = \det(sI - A) = s^3$

$\Rightarrow p(A) = A^3 = 0$.

$n \geq 3 \quad A^n = 0$

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Minimal polynomial of A

The minimal polynomial (minpoly) of a square matrix A is the monic polynomial q of lowest order for which $q(A) = 0$.

The order of minpoly q is always less than or equal to the order of charpoly p; its roots are precisely the eigenvalues of A and $p(s) = g(s)q(s)$.

Example 88 The matrix

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$p(s) = s^2$ $\quad q(s) = s$

$2 \geq 1$ ✓

$\times q(s) = s \Rightarrow q(A) = A \neq 0$

$\checkmark q(s) = s^2 \Rightarrow q(A) = 0$

$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ $\quad p(s) = s^2$ $\quad q(s) = s^2$

$2 \geq 2$ ✓

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Exercise 90 Without doing any matrix multiplications, compute A^4 for

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 564 & 1 & 0 & 0 \end{pmatrix}$$

HW

→ Companion matrix ✗

$p(\cdot) = \det(sI - A) = \det\begin{pmatrix} s & -1 & \cdots & 0 \\ 0 & s & & \\ \vdots & & \ddots & \end{pmatrix}$

$p(\cdot)$ can be easily solved

$\Rightarrow p(A) \Rightarrow A^n$

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The matrix exponential

$e^A = ?$, A is a matrix

e^a = 0 值

for $k = 1, 2, \ldots$; hence for any complex number λ,

$$e^{\lambda} = \sum_{k=0}^{\infty} \frac{1}{k!}\lambda^k = 1 + \lambda + \frac{1}{2}\lambda^2 + \frac{1}{3!}\lambda^3 + \ldots$$

$p(\lambda) =$ λ^0 λ^n $n=\infty$

Thus the exponential of any square matrix A is defined by:

$$e^A := \sum_{k=0}^{\infty} \frac{1}{k!}A^k = I + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \cdots$$

$p(A) =$

this power series coverages for every square matrix A

(a) For any zero matrix, $A = 0$

$e^0 = I$

(b) Suppose t is a scalar variable

$$\frac{de^{At}}{dt} = Ae^{At} = e^{At}A$$

$Ap(A) = p(A)A$ (a)

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The following results also hold: (A is a square matrix)

① $e^{A(t_1+t_2)} = e^{At_1} e^{At_2}$

$e^{-A} = (e^{A})^{-1}$

② If λ is an eigenvalue of A w/ eigenvector v, then e^{λ} is an eigevalue of e^{A} w/ "

③ e^{A} is nonsingular

④ $e^{A+B} = e^{A} e^{B}$ if A, B commute

$= e^{B} e^{A}$

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The state transition matrix

$\dot{x} = Ax, \quad x(0) = x_0$

$x(t) = e^{At} x_0$

Using the above definitions and abusing notation (e^{At} instead of e^{tA}),

$$e^{At} := \sum_{k=0}^{\infty} \frac{1}{k!}(tA)^k = I + tA + \frac{1}{2!}(tA)^2 + \frac{1}{3!}(tA)^3 + \dots$$

① Matlab/ python : expm(·)

② numerical simulation

$\Phi(t) = e^{At} = [\phi_1(t) \ \phi_2(t) \ \dots \ \phi_n(t)]$

$x(t) = \Phi(t) x_0 = x_{0,1}\phi_1(t) + x_{0,2}\phi_2(t) + \dots + x_{0,n}\phi_n(t)$

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③ Jordan form

$e^{At} = T e^{\Lambda t} T^{-1}$ $(A = T \Lambda T^{-1})$

Λ is the Jordan form of A

④ Laplace

$$\mathcal{L}(e^{At}) = (sI - A)^{-1}$$

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$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} \quad e^{At} = ?$$

So,

$$(sI - A)^{-1} = \begin{pmatrix} \frac{s-3}{(s-2)(s-4)} & \frac{-1}{(s-2)(s-4)} \\ \frac{-1}{(s-2)(s-4)} & \frac{s-3}{(s-2)(s-4)} \end{pmatrix}$$

$$\frac{s-3}{(s-2)(s-4)} = \frac{1}{2}\left(\frac{1}{s-2} + \frac{1}{s-4}\right)$$

$$\frac{-1}{(s-2)(s-4)} = \frac{1}{2}\left(\frac{1}{s-2} - \frac{1}{s-4}\right)$$

$$\mathcal{L}^{-1}\left(\frac{1}{s-2}\right) = e^{2t} \qquad \mathcal{L}^{-1}\left(\frac{1}{s-4}\right) = e^{4t}$$

$$e^{At} = \mathcal{L}^{-1}\left((sI-A)^{-1}\right) = \frac{1}{2}\begin{pmatrix} e^{2t}+e^{4t} & e^{2t}-e^{4t} \\ e^{2t}-e^{4t} & e^{2t}+e^{4t} \end{pmatrix}$$

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An explicit formula for $(sI - A)^{-1}$

Recall that

$$\boxed{(sI - A)^{-1} = \frac{1}{d(s)} N(s)} \tag{7.6}$$

where d, the characteristic polynomial of A, can be expressed as

$$d(s) = s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0$$

and N, an $n \times n$ matrix of polynomials of order less than n, can be expressed as

$$N(s) = s^{n-1}N_{n-1} + s^{n-2}N_{n-2} + \cdots + sN_1 + N_0$$

→ proof Caley-Hamilton

scalar

Hence

$(sI-A)^{-1} = \frac{1}{d(s)} \Rightarrow a_0, a_1, \cdots, a_{n-1},$

$N_0, N_1, \cdots, N_{n-1}$

Assume the scalar $a_0, \cdots a_{n-1}$ have been computed

$N(s) = d(s)(sI-A)^{-1}$

Recall

$(sI-A)^{-1} = \frac{1}{s}I + \frac{1}{s^2}A + \frac{1}{s^3}A^2 + \cdots$

$(s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0)\left(\frac{1}{s}I + \frac{1}{s^2}A + \frac{1}{s^3}A^2 + \cdots\right)$

Equating coefficients of power of s ...

s^{n-1}: $N_{n-1} = I$

s^{n-2}: $N_{n-2} = A + a_{n-1}I$

s^{n-3}: $N_{n-3} = A^2 + a_{n-1}A + a_{n-2}I$

$\vdots$

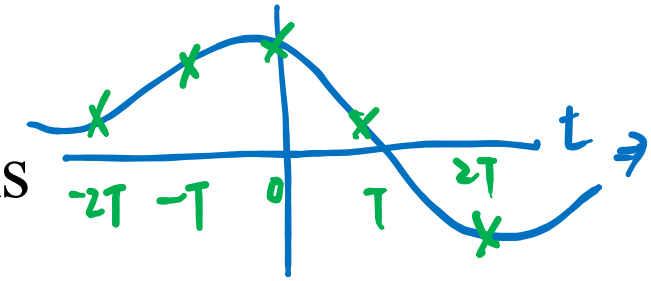
s^0: $N_0 = A^{n-1} + a_{n-1}A^{n-2} + \cdots + a_2 A + a_1 I$

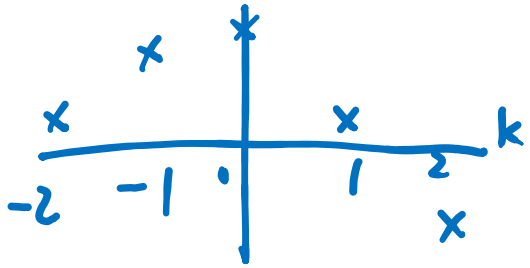
s^{-1}: $0 = A^n + a_{n-1}A^{n-1} + a_{n-2}A^{n-2} + \cdots + a_1 A + a_0 I$

$\Rightarrow$ proof of Caley-Hamilton

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Sampled-data systems





Consider a continuous-time LTI system described by

$$\dot{x}_c(t) = A_c x_c(t)\,.$$

Suppose we only look at the behavior of this system at discrete instants of time, namely

$$\cdots, -2T, T, 0, T, 2T, \cdots$$

where $T > 0$ is called the sampling time.

$x_d(k) = x_c(kT)$, $k = \cdots, -2, -1, 0, 1, 2, \cdots$

$\Rightarrow x_d(k+1) = x_c((k+1)T)$ $\quad x(t) = e^{A(t-t_0)} x(t_0)$

$= e^{A_c T} x_c(kT)$

$= \boxed{e^{A_c T}} x_d(k) = \boxed{A_d} x_d(k)$

$A_d = e^{A_c T}$

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$\lambda_1, \lambda_2, \cdots \lambda_i$ are eigenvalues of A_c

$\Rightarrow e^{\lambda_1 T}, e^{\lambda_2 T}, \cdots, e^{\lambda_i T}$ are eigenvalues of $e^{A_c T}$ (A_d)

$|e^{\lambda_i T}| < 1 \iff \mathbb{R}(\lambda_i) < 0$

$|e^{\lambda_i T}| = 1 \iff \mathbb{R}(\lambda_i) = 0$

$|e^{\lambda_i T}| > 1 \iff \mathbb{R}(\lambda_i) > 0$

$e^{\lambda_i T} = 1 \iff \lambda_i = 0$

If we approximate $e^{A_c T}$ by two terms in its power series

$\Rightarrow A_d = e^{A_c T} \approx I + A_c T$

* all eigenvalues of A_d are nonzero $\Rightarrow A_d$ is invertible

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Behavior of (LTI) systems

- Consider now the unattached mass

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= 0 \end{aligned}$$

With initial state $x(0) = (0,\ 1)$, this system has solution $x(t) = (t,\ 1)$. The term t cannot be expressed as a linear combination of exponential functions. So, the system matrix

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$ is not diagonalizable

$p(s) = s^2$ ⇒ 只有 1 個 eigenvalue (2×2) ⇒ All eigenvectors are $v = \begin{pmatrix} x_1 \\ 0 \end{pmatrix}$

⇒ 只有 1 個 linear independent eigenvector

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Jordan blocks

$A = n \times n$, n 個 distinct eigenvalues

complex Jordan form $\begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}$ $\lambda_i \in \mathbb{C}$

For any complex number λ and any positive integer n, $J(\lambda, n)$ is a **Jordan block** of order n; it is an $n \times n$ matrix with every diagonal element equal to λ, every element of the superdiagonal equal to 1 and every other element equal to 0, that is, it has the following structure:

$$J(\lambda, n) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ \vdots & & & & & \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{pmatrix}$$

Lighted mass $\dot{x}_1 = x_2$

$\dot{x}_2 = 0$

$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$

$p(s) = s^2 \quad \lambda = 0$

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Example 93 (The Disturbed unattached mass)

$$\ddot{q} = w$$

The forcing term w is a constant disturbance input. Introducing state variables

$$x_1 := q \qquad x_2 := \dot{q} \qquad x_3 = w$$

we obtain

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= 0 \end{aligned}$$

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \lambda_1 = \lambda_2 = \lambda_3 = 0$$

$$p(s) = \det(sI - A) = s^3$$

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The characteristic polynomial of $J(\lambda, n)$ is given by

$$p(s) = (s - \lambda)^n.$$

Companion matrix w/ last row is zero

$$J(\lambda,n) - \lambda I = J(0,n) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & & & & & \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}$$

$v = 0$

all eigenvectors of $J(\lambda, n)$ are

$$v = \begin{pmatrix} v_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Hence λ : AM = n, GM = 1

characteristic poly

$$p(s) = \det(sI - (J(\lambda,n) - \lambda I)) = s^n$$

We now have

$$\det(sI - J(\lambda,n)) = \det(sI - (J(\lambda,n) - \lambda I) - \lambda I)$$
$$= \det((s-\lambda)I - (J(\lambda,n) - \lambda I))$$
$$= \det(zI - (J(\lambda,n) - \lambda I))$$
$$= z^n = (s-\lambda)^n$$

$z = s - \lambda$

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Fact 11 *If* $\Lambda = J(\lambda, n)$, *then*

$$f(\Lambda) = \begin{pmatrix} f(\lambda) & f'(\lambda) & f''(\lambda)/2! & \dots & f^{(n-2)}(\lambda)/(n-2)! & f^{(n-1)}(\lambda)/(n-1)! \\ 0 & f(\lambda) & f'(\lambda) & \dots & f^{(n-3)}(\lambda)/(n-3)! & f^{(n-2)}(\lambda)/(n-2)! \\ \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & f(\lambda) & f'(\lambda) \\ 0 & 0 & 0 & \dots & 0 & f(\lambda) \end{pmatrix}$$

Hence,

$$f(\Lambda) = e^{\Lambda t} = \begin{pmatrix} e^{\lambda t} & te^{\lambda t} & t^2 e^{\lambda t}/2! & \cdots & t^{n-2}e^{\lambda t}/(n-2)! & t^{n-1}e^{\lambda t}/(n-1)! \\ 0 & e^{\lambda t} & te^{\lambda t} & & & \\ \vdots & & & & & \\ 0 & 0 & 0 & \cdots & e^{\lambda t} & te^{\lambda t} \\ 0 & 0 & 0 & \cdots & 0 & e^{\lambda t} \end{pmatrix}$$

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Generalized eigenvectors

A is defective, $(A-\lambda I)g = v \neq 0$, $(A-\lambda I)^2 g = 0$ → λ: GM > 2, GM = 1

DEFN. A non-zero vector $g^k \in \mathbb{C}^n$ is a generalized eigenvector of order k of A if it satisfies

k是次方

$$\boxed{\begin{aligned}(A-\lambda I)^{k-1} g^k &\neq 0 \\ (A-\lambda I)^k g^k &= 0\end{aligned}}$$

不是次方 $g^2 \neq g \cdot g$.

for some complex number λ.

$k=1$ $(A-\lambda I)g^1 = 0 \Rightarrow g^1 = v$

an eigenvector is a generalized eigenvector of order 1

* if g^k, order k $\Rightarrow v = (A-\lambda I)^{k-1} g^k$

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Example 94

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = J(0,2)$$

Here $\lambda = 0$ is the only eigenvalue. For

$\hookrightarrow V = g^1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. generalized e-vector of order 1

$(A-\lambda I)\, g^1 = 0$

$(A-\lambda I)^2 g^2 = 0$, $(A-\lambda I) g^2 = g^1$

$\hookrightarrow g^2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow A g^2 = g^1 \neq 0$

$A^2 g^2 = A g^1 = 0$

4 order 2

If A is a Jordan block

$\bigstar\ g^k := \begin{pmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \leftarrow k^{th}$ element

is a generalized e-vector of order k

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Chain of generalized eigenvectors

$v = g^1 \quad (A-\lambda I)g^1 = 0 \quad (A-\lambda I)g^2 = g^1 \neq 0$

$(A-\lambda I)^2 g^2 = 0$

It follows from the above definition that if g^k is a generalized eigenvector of order $k > 1$ then,

$$g^{k-1} := (A - \lambda I)g^k \neq 0$$

is a generalized eigenvector of order $k - 1$. To see this:

$(A-\lambda I)^{k-1} g^{k-1} = (A-\lambda I)^k g^k = 0$

$(A-\lambda I)^{k-2} g^{k-1} = (A-\lambda I)^{k-1} g^k \neq 0$

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Consider any generalized e-vector g^k of order $k > 1$

$g^1, g^2, \cdots, g^k$

$$g^{k-1} := (A-\lambda I)\, g^k$$

$$g^{k-2} := (A-\lambda I)\, g^{k-1} = (A-\lambda I)^2 g^k$$

$$\vdots$$

$$g^1 := (A-\lambda I)\, g^2 = (A-\lambda I)^{k-1} g^k$$

$\Rightarrow$ This is called "a chain of k generalized e-vector)

If A is a Jordan block $J(\lambda, n)$ of length n.

$g^1, g^2, \cdots g^n$ is a chain of generalized e-vector

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The Jordan form of a square matrix

We consider first the special case in which the $n \times n$ matrix A has only a single eigenvalue λ and this eigenvalue has geometric multiplicity one.

Hence the algebraic multiplicity of λ is n and the charpoly of A is

$$p(s) = (s - \lambda)^n$$

If A is diagonalizable

$T = (v_1, v_2, v_3 \cdots v_n)$

Similarity transformation

$\Lambda = T^{-1} A T$

Complex Jordan form of A

Let $g^1 = v$, $g^1 \neq 0 \Rightarrow Ag^1 = \lambda g^1$ $(Av = \lambda v)$

Recursively solve

$(A - \lambda I) g^2 = g^1$

$\vdots$

$(A - \lambda I) g^n = g^{n-1}$

$\Rightarrow$

$Ag^1 = \lambda g^1$

$Ag^2 = g^1 + \lambda g^2$

$Ag^3 = g^2 + \lambda g^3$

$\vdots$

$Ag^n = g^{n-1} + \lambda g^n$

$T := (g_1\ g_2 \cdots g_n)$

$\Lambda := J(\lambda, n)$

The chain is a basis for $\mathbb{C}^n$

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Λ (Jordan block)

Example 95

$$A = \begin{pmatrix} -3.5 & -0.5 \\ 0.5 & -4.5 \end{pmatrix}$$

Characteristic polynomial of A:

$$p(s) = \det(sI - A) = (s+4)^2.$$

$\lambda = -4 \Rightarrow AM = 2 = n$, $GM = 1$

So, A has a single eigenvalue $\lambda = -4$. Solving $(A - \lambda I)v = 0$ yields one linearly independent eigenvector

$$v = g^1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$(A - \lambda I)\, g^2 = g^1$$

$$g^2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\Lambda = T^{-1} A T, \quad T = (g_1 \; g_2) = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\Lambda = T^{-1} A T = \begin{pmatrix} -4 & 1 \\ 0 & -4 \end{pmatrix} = J(-4, 2)$$

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Block-diagonal matrices

Suppose $A_1, A_2, \ldots, A_l$ are l matrices (not necessarily square). By

$$A = \text{blockdiag}\,(A_1,\ A_2,\ \ldots, A_l)$$

we mean

$$A = \begin{pmatrix} A_1 & 0 & \ldots & 0 \\ 0 & A_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & 0 & \ldots & A_l \end{pmatrix}$$

$A^{-1} = \text{blockdiag}\,(A_1^{-1}, \cdots, A_l^{-1})$

$\det A = \det A_1 \det A_2 \cdots \det A_l$

$\det(sI - A) = \det(sI - A_1) \cdots \det(sI - A_l)$

$\text{rank}\, A = \text{rank}\, A_1 + \cdots + \text{rank}\, A_l$

$\text{nullity}\, A = \text{nullity}\, A_1 + \cdots + \text{nullity}\, A_l$

① $\alpha A = \text{blockdiag}\,(\alpha A_1, \alpha A_2, \cdots \alpha A_l)$

② If $B = \text{blockdiag}\,(B_1, B_2, \cdots, B_l)$

$A + B = \text{blockdiag}\,(A_1 + B_1, A_2 + B_2, \cdots, A_l + B_l)$

$AB = \text{blockdiag}\,(A_1 B_1, A_2 B_2, \cdots A_l B_l)$

③ If $f(\cdot)$ is an analytic func.

$f(A) = \text{blockdiag}\,(f(A_1), \cdots f(A_l))$

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$$A = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 1 & 0 \\ 0 & 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 7 \end{pmatrix} = \begin{pmatrix} J(2,3) & & \\ & J(5,2) & \\ & & J(7,1) \end{pmatrix} = \text{blocking}(J(2,3), J(5,2), J(7,1))$$

$$\det(sI - A) = \det(sI - J(2,3)) \cdot \det(sI - J(5,2)) \cdot \det(s-7)$$

$$= (s-2)^3 (s-5)^2 (s-7)$$

A has 3 e-values, $\lambda_1 = 2$, $\lambda_2 = 5$, $\lambda_3 = 7$

$\lambda_1 = 2$: $v_1 = g^1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ g^2, g^3

$\lambda_2 = 5$: $v_2 = g^1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$ g^2 ← 1

$\lambda_3 = 7$: $v_3 = g^1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$

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$$A = \left(\begin{array}{ccc|cc|c} 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 2 \end{array}\right) = \text{block diag}\left[\underbrace{J(2,3)}_{A_1}, \underbrace{J(2,2)}_{A_2}, \underbrace{J(2,1)}_{A_3}\right)$$

The "index" of λ of a matrix A is the smallest integer k such that

$$\text{nullity}((A-\lambda I)^{k+1}) = \text{nullity}((A-\lambda I)^{k})$$

① 最大 Jordan block 大小

② 最大 chain 的長度

$$\det(sI-A) = (s-2)^3 \cdot (s-2)^2 \quad (s-2)^1$$

$$= (s-2)^6$$

A has only one single e-values $\lambda = 2$, $AM = 6$

but A has at most 3 linear independent e-vectors ($GM = 3$)

A_1: $\underline{V} = g^1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ A_2: $\underline{V} = g^1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$ g^2 A_3: $\underline{V} = g^1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$ g^3

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$$A = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 5 \end{pmatrix} = \text{blockdiag}\,(J(2,2), J(2,2), J(5,2), J(5,1))$$

$\det(sI - A) = (s-2)^4 (s-5)^3$

$\lambda = 2$ AM = 4, GM = 2 ⇒ A 有 4 個 linear independent e-vectors

$\lambda = 2$

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$\lambda = 5$ AM = 3 GM = 2

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

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Fact 12 *If q is the geometric multiplicy of an eigenvalue λ, then there is a basis consisting of q chains of generalized eigenvectors.*

$$\begin{matrix} (A-\lambda I)^{k_1-1}g^{k_1}, & \ldots, & g^{k_1} \\ \vdots & & \\ (A-\Lambda I)^{k_q-1}g^{k_q}, & \ldots, & g^{k_q} \end{matrix}$$

where $k_1 \geq k_2 \geq \ldots k_q$ and k_1 is the index of λ.

Fact 13 *For every square matrix there is a* basis of generalized eigenvectors. *This basis consists of chains of generalized eigenvectors. For a specific eigenvalue, the number of chains associated with it equals its geometric multiplicity, and the the length of the largest chain associated with it equals its index.*

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LTI systems

Recall that the solution to

$$\dot{x} = Ax, \qquad x(0) = x_0$$

is given by

$$x(t) = e^{At}x_0 \quad (= Ad)$$

Suppose $\lambda_1, \cdots \lambda_l$ are evalues of A & n_i is the index of λ_i

$$A = T \Lambda T^{-1}$$

$$\Lambda = \text{blockdiag}(J_1, J_2, \cdots, J_l)$$

$$e^{At} = T\ \text{blockdiag}(e^{J_1 t}, e^{J_2 t}, \cdots e^{J_l t}) T^{-1}$$

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Input-output systems

$\dot{x} = Ax$

$x(t) = e^{At} x_0$

In general, a state space description of an LTI (linear time-invariant) continuous-time system with inputs and outputs is given by

$$\begin{array}{rcl} \dot{x} & = & Ax + Bu \\ y & = & Cx + Du \end{array}$$

y ← u ←

$X(t) = ?$

sys dynamics

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State response and the convolution integral

The solution to

$$\dot{x}(t) = Ax(t) + Bu(t) \qquad \text{with} \qquad x(t_0) = x_0$$

is unique and is given by

$$x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^{t} e^{A(t-\tau)}Bu(\tau)\,d\tau$$

for all t

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Proof

Consider any time t, τ

τ in the interval bound t_0 & t

$$\dot{X}(\tau) = AX(\tau) + Bu(\tau)$$

① $\times e^{-A\tau}$

$$e^{-A\tau}\dot{X}(\tau) - e^{-A\tau}AX(\tau) = e^{-A\tau}Bu(\tau)$$

$$\frac{d}{d\tau}\left(e^{-A\tau}X(\tau)\right) = e^{-A\tau}Bu(\tau)$$

② Integrate t_0 to t ! $X(t_0) = X_0$

$$e^{-At}X(t) - e^{-At_0}X_0 = \int_{t_0}^{t} e^{-A\tau}Bu(\tau)\,d\tau$$

③ $\times e^{At}$

$$X(t) - e^{A(t-t_0)}X_0 = \int_{t_0}^{t} e^{-A\tau}Bu(\tau)\,d\tau$$

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* $x^p(\cdot)$ is a particular sol, $x^p(t_0) = x_0^p$

$$x(t) = e^{A(t-t_0)}(x_0 - x_0^p) + x^p(t)$$

微分 ↓

$$\dot{x}(t) = Ae^{A(t-t_0)}(x_0 - x_0^p) + \dot{x}^p(t)$$

$$= A\left(e^{A(t-t_0)}(x_0 - x_0^p) + x^p(t)\right) + Bu(t)$$

$$= Ax(t) + Bu(t)$$

$\dot{x} = Ax + Bu$

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Impulse response

$\dot{x} = Ax + Bu$

$= Ax + B\delta(t)v$

常數向量 (決定 impulse 大小與方向)

$\int \delta(t) = 1, \ \delta(t) = 0 \quad t \neq t_0$

Suppose $t_0 = 0$ and

$$u(t) = \delta(t)v$$

where $\delta(\cdot)$ is the Dirac delta function or the unit impulse function. (Don't confuse with the δ used in linearization.) Recall that if $[t_0, t]$ is a time interval containing zero and g is any continuous function then,

$$\Rightarrow \int_{t_0}^{t} g(t)\delta(t)dt = g(0)$$

the sol of sys :

$$X(t) = e^{A(t-t_0)}X_0 + e^{A(t-t_0)}BV$$

zero initial state response ($X_0 = 0$)

$$X(t) = e^{A(t-t_0)}BV$$

$\Rightarrow$ this is same as the sys response to zero input & initial state $X_0 = BV$

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Response to exponentials

Suppose $t_0 = 0$ and

$$u(t) = e^{\lambda t} v$$

where v is a constant m-vector and λ is not an eigenvalue of A. Then the solution

$$x(t) = \underbrace{e^{At}(x_0 - x_0^p)}_{\text{transient part}} + \underbrace{e^{\lambda t} x_0^p}_{\text{steady state part}}$$

where

$$x_0^p = (\lambda I - A)^{-1} B v$$

when A is asymptotically stable, $t \to \infty$, $\Rightarrow 0$

$x_0 = x_0^p$ 的推導 the result of

$$\dot{x} = Ax + Bu$$

Consider $x(t) = e^{\lambda t}(\lambda I - A)^{-1} B v = x_0^p$

$\dot{x}(t) = \lambda e^{\lambda t}(\lambda I - A) B v = \lambda x(t)$

如果 x 的是 eigenvector $\Rightarrow Ax$

$(\lambda I - A) x(t) = e^{\lambda t} B v = B u(t)$

$\lambda x(t) = Ax(t) + B u(t)$

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Output response

$$x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^{t} e^{A\tau} B u(t)\, d\tau$$

The output response of the system

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \end{aligned}$$

subject to initial condition $x(t_0) = x_0$ is unique and is given by

$$y(t) = C e^{A(t-t_0)}x_0 + \int_{t_0}^{t} G(t-\tau)u(\tau)\, d\tau$$

when $u(t) = f(t)v$

$$G(t) = C e^{A(t-t_0)}B + f(t)Dv$$

從 x(t) 來的

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To see this, recall that if zero is in the interval bounded by t_0 and t and g is any continuous function then,

$$\int_{t_0}^{t} \delta(t) g(t) dt = g(0)$$

Recalling that

$$x(t) = e^{A(t-t_0)} x_0 + \int_{t_0}^{t} e^{A(t-\tau)} B u(\tau) d\tau$$

we see that

$$y(t) = Cx(t) + Du(t)$$

$$= C e^{A(t-t_0)} x_0 + C \int_{t_0}^{t} e^{A(t-\tau)} B u(\tau) + \delta(t-\tau) D u(\tau) \, d\tau$$

$$= C e^{A(t-t_0)} x_0 + \int_{t_0}^{t} G(t-\tau) u(\tau) \, d\tau$$

$$\Rightarrow G(t-\tau) = e^{A(t-t_0)} B D + \delta(t) D$$

zero initial state $x(0) = 0$, $t_0 = 0$

$$y(t) = \int_{0}^{t} G(t-\tau) u(\tau) \, d\tau \qquad u \rightarrow \boxed{G} \rightarrow y$$

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Example 103

$$\begin{aligned} \dot{x} &= -x + u \\ y &= x \end{aligned}$$

$u = f(t)$

$A = -1 \quad B = 1$

$C = 1 \quad D = 0$

$$G(t) = C e^{At} B + f(t) D$$

$$= e^{-t}$$

for zero initial state $x(0) = 0$

$$y(t) = \int_0^t G(t-\tau)\, u(\tau)\, d\tau$$

$$= \int_0^t e^{-(t-\tau)} f(\tau)\, d\tau$$

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Example 104 Unattached mass

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= u \\ y &= x_1 \end{aligned}$$

Here

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad C = \begin{pmatrix} 1 & 0 \end{pmatrix} \qquad D = 0$$

Hence

$$e^{At} = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

impulse response

$$G(t) = Ce^{At}B$$

$$= t$$

$$y(t) = \int_0^t (t-\tau) f(\tau)\, d\tau = \text{——}$$

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* The TF of an LTI is the Laplace transform of its impulse response matrix

$$\hat{G} = \mathcal{L}(G(t))$$

Transfer function and impulse response. Consider the impulse response matrix given by

$$G(t) = Ce^{At}B + \delta(t)D.$$

impulse func.

Using the following Laplace transform results:

$$\mathcal{L}(e^{At})(s) = (sI - A)^{-1}, \qquad \mathcal{L}(\delta) = 1$$

we obtain

$$\mathcal{L}(G(t)) = \underbrace{C(sI-A)^{-1}B + D}_{\hat{G}}$$

$\hat{G}$ transfer function

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Response to exponentials ~~and sinusoids~~

Suppose $t_0 = 0$ and

$$u(t) = e^{\lambda t} v$$

where v is a constant m vector and λ is not an eigenvalue of A. Recall that

$$x(t) = e^{At}(x_0 - x_0^p) + e^{\lambda t} x_0^p$$

where $x_0^p = (\lambda I - A)^{-1} B v$ Since $y(t) = Cx(t) + Du(t)$, we obtain that

sys's TF evaluated at λ

$$y(t) = \left(e^{At}(x_0 - x_0^p) \right) + e^{\lambda t} \hat{G}(\lambda) v$$

$$\hat{G}(\lambda) = C(\lambda I - A)^{-1} B + D$$

If A is AS, transient part goes to 0 as $t \to \infty$

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